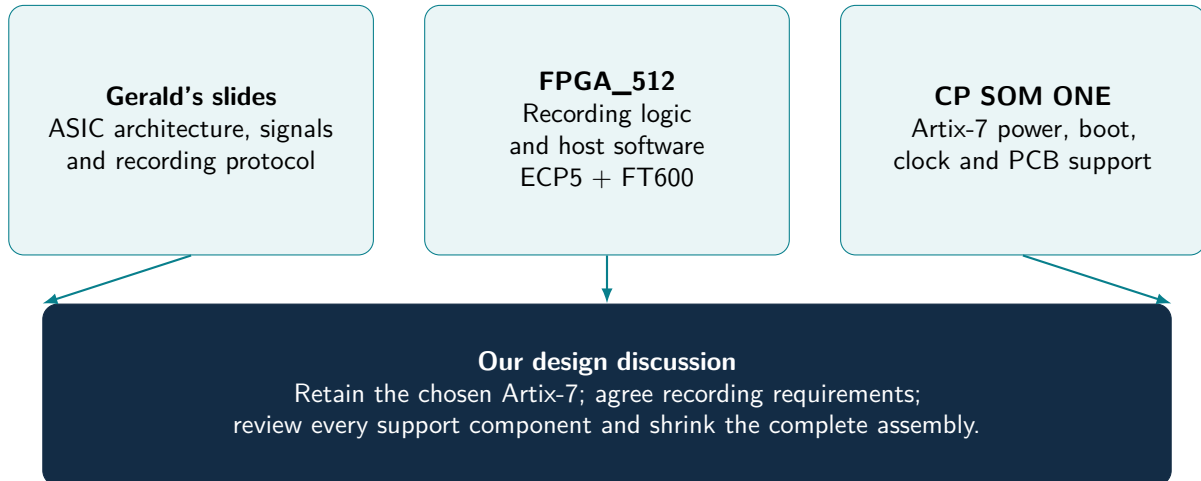


01 / THE CONNECTION

Three sources for the recording PCB.

Keep the chosen CP SOM ONE FPGA. Use FPGA_512 to understand the recording job; use CP SOM ONE as the starting support circuit. Adapt what is needed instead of starting from an empty schematic.



The important distinction

These projects use **different FPGA families**. Recording behavior may be reusable, but the old bitstream, pin numbers, clock primitive and device build cannot be copied directly to Artix-7. CP SOM ONE does not already contain the complete FPGA_512 recorder. [A–D]

How this packet treats answers

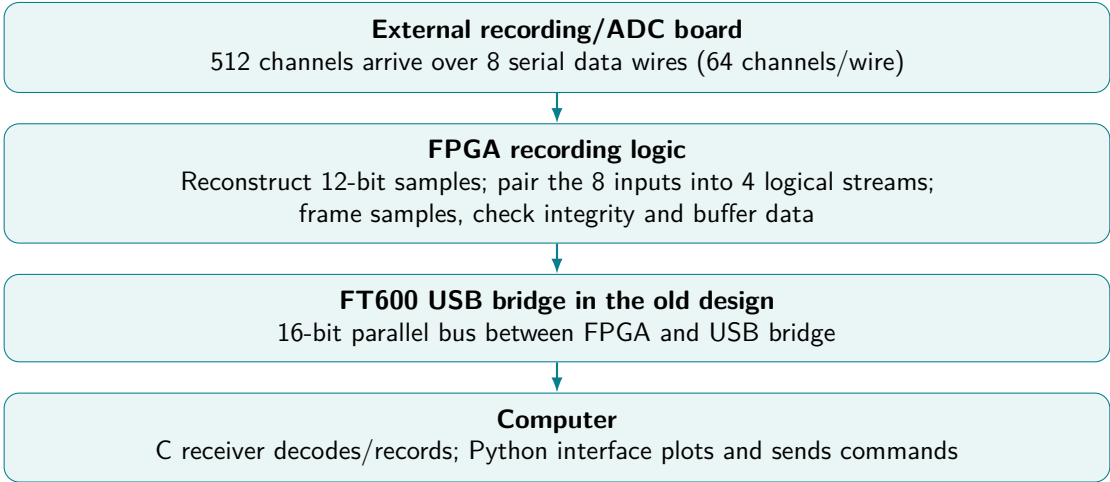
- **Found:** directly present in a repository, schematic or constraint file. This is source evidence, not a measurement on the lab board.
- **Confirm:** a useful starting value or function from the reference; Gerald must confirm it applies to the new design.
- **Open:** the files do not settle it. Keep it TBD until an answer or document is supplied.

Read pages 1–3 first. Pages 4–6 explain what the PCB already provides and what may change. Pages 7–9 compare 4K scenarios, list remaining questions and provide the change worksheet. Page 10 gives sources. Pages 11–14 connect Gerald's slides and explain startup and shrink constraints. Pages 15–23 give exact parts and questions. **Pages 24–26 show every capacitor on the PCB and explain its job.**

Working target from Howard: about 4K channels. One FPGA versus two FPGAs is still open. We use the 512-channel implementation as the building block and compare conditional 4,096 and $2 \times 2,048$ channel allocations on page 7.

Revision 5, 16 September 2026. Supersedes the previous meeting print copy. No PCB redesign, component replacement or hardware test is claimed.

What carries over from FPGA_512?



Three different paths: acquisition data goes toward the computer; runtime commands return through FT600 to the FPGA; FPGA programming uses JTAG/flash. The checked acquisition path does not show a separate MCU. [A,B]

Part of old system	Useful starting point	Work for the Artix board
Sample/framing logic	Channel ordering, 12-bit samples, tagged words, CRC and sequence checks.	Port and verify logic; implement timing/clock-domain crossings on Artix.
Clock and FPGA memory	Nominal 16 MHz board clock, returned capture clock, on-chip FIFOs.	Replace ECP5 clock primitive; verify Artix RAM inference, timing and reset behavior.
USB + host software	FT600 transport, decoder, plotting and WAV recording.	Retain/adapt if transport and frame format match; another bridge changes transport work.
Board pin assignments	An inventory of needed signals.	Create a fresh Artix pin/bank plan and timing constraints; old package pins do not transfer.

8 wires, 4 streams, 512 channels describe different things. The eight physical ADC wires are paired into four framed streams. They are not four recording channels.

Main scope: adc_4lane_framed. It also has a four-wire serial mirror plus clock; separate UWB test trees exist. Which output path the team needs remains open. [A,B]

03 / PREFILLED REQUIREMENTS

What we can answer now

The middle column records source evidence. It does not silently become an approved requirement for the new board.

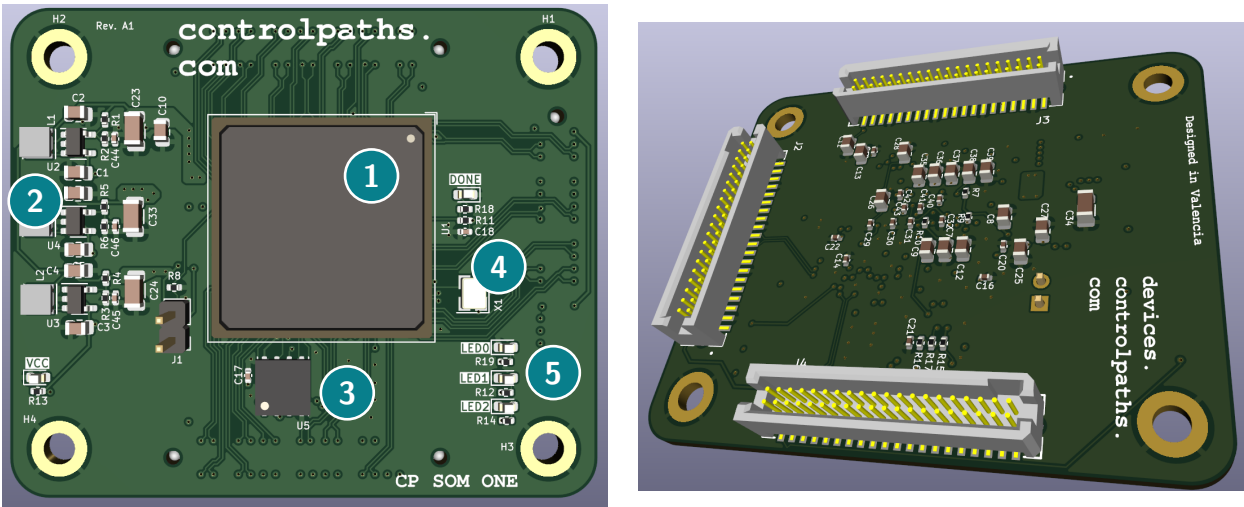
Question	Found in the references	For our board
Channel count	512 = 8 wires × 64 channels. [A]	Confirm 4K target, exact total and partition.
Sample width	12-bit ADC sample; 16-bit tagged transport words. Host recording uses 16-bit PCM. [A,B]	Confirm retain format and processing.
Sample rate	Host: 15,625 samples/s/channel. Gerald's slides: 31,250; 1,024 clocks/frame. [A,B,H]	Conflict: confirm actual/required mode.
FPGA job	Deserialize, align/interleave, frame, CRC/sequence, buffer; reset/configuration/stimulation controls. [B]	Confirm which functions remain.
Output destination	FT600 USB to computer daemon, live plots and WAV files. [A,B]	Confirm USB, serial/UWB or other route.
MCU role	No separate MCU in the examined main acquisition/control path. [A,B]	Open MCU identity and responsibilities.
Signal levels	ASIC pins: LVCMOS15 (1.5 V). FT600 pins: LVCMOS25 (2.5 V). Serial mirror: 3.3 V. [C]	Confirm actual hardware levels; redesign bank plan.
Clocking	16 MHz board input forwarded to ASIC; returned clock used for capture; 100 MHz FT600 constraint. [B,C]	Confirm actual returned timing and required clocks.
Buffering	32,768 × 16-bit main FIFO = 64 KiB, plus smaller FIFOs. [B]	Confirm allowable pauses and Artix memory allocation.
Data loss policy	Overflow/drop, CRC and sequence tracking exist; host can zero-fill missing samples. [A,B]	Open accepted loss, fault reporting and duration.
FPGA boot	Build has JTAG programming and flash targets for ECP5. [D]	Confirm retain CP flash or redesign MCU boot.
Size / power / temperature	Not established as new-board limits by this source set.	Open numerical limits and owner.

Data continuity on a plot is not proof of lossless capture. Error counters and any inserted samples must be considered when agreeing what “successful recording” means.

Source statements are checked at pinned commits on page 10. Exact source locations and caveats are in `research/fpga512_connection_audit.md`. No bench behavior has been verified in this comparison.

04 / LOOK AT THE STARTING PCB

What CP SOM ONE already provides



GitHub renders: top side left, bottom side right. Pablo Trujillo / ControlPaths. Labels added here; image license on page 10. [E]

Existing function		What it gives us
1	Artix-7 FPGA	The chosen chip stays. Author populated 75T using a 35T-compatible schematic; record the exact lab ordering code.
2	Three power circuits	Convert +5 V input into 1.0 V, 1.8 V and 3.3 V rails. Actual load/ramp suitability still needs review.
3	Boot flash	128 Mbit = 16 MiB of nonvolatile storage for FPGA configuration. This is not a substitute for a fast recording buffer.
4	Oscillator	An external timing source. Author states 30 MHz; BOM lacks the full programmed frequency suffix.
5	Status lights	Five LEDs total: configuration DONE, input power and three user LEDs.
Other side / headers		Local power capacitors, three 40-contact board headers, JTAG access and mounting holes.

It is an FPGA support module, not a complete recorder. The populated design has no recording ADC/front end, MCU, FT600/USB interface or external working RAM. Those functions must be supplied elsewhere or added if required.

Nominal reference PCB: 60 × 50 mm, four copper layers, 1.6 mm thick. Those dimensions are reference choices, not the new target. Sources: KiCad PCB, netlist and BOM [E]; author explanation [F].

05 / START FROM THE EXISTING BOM

A list you can add to or subtract from

79 listed component placements, grouped by function below. Four connector footprints are additional and missing from the production BOM. Counts help auditing; they do not predict saved board area. [E]

Existing block	Parts / count	Meeting choice and condition
FPGA	U1: 1	Keep exact part. No device/package substitution in this task.
Power generation	U2–U4, L1–L3, R1–R6, C1–C6, C44–C46: 21	Keep, replace or move generation. Required rails/current/stability remain; evaluate full circuit area.
Rail bulk/bypass	C7–C43: 37	Keep a validated local power network. Recalculate only with electrical justification; do not delete by count.
Boot storage	U5 + R15–R17: 4	Keep flash; change compatible part; or remove only if another device reliably configures each boot.
Clock	X1: 1	Keep/change oscillator or use a suitable external source. Resolve 30 MHz versus old 16 MHz clock plan.
Startup settings	R7–R11: 5	Preserve defined configuration behavior. Resistor values/connections require a reviewed design.
LEDs + resistors	D1–D5 + R12–R14, R18–R19: 10	Possible removal/reduction if status/test access is retained elsewhere. Keep R11 pending configuration review.

Connectors and mechanics – often a larger area opportunity

- **J2–J4: three 40-contact headers, 1.27 mm pitch.** They expose 76 signal wires plus four JTAG wires; remaining contacts include power/grounds. These are not 120 free GPIO pins.
- **J1: two-pin boot-mode selector.** Can potentially become fixed straps/solder pads, with a defined recovery method.
- **H1–H4: four mounting holes.** Revise only after agreeing support, enclosure and assembly requirements.

Two power-circuit checks before copying it: U3/U4 supplier codes disagree with their BOM names. C44–C46 connect regulator feedback to ground, not FPGA supply to ground; review them against the actual regulator’s stable reference circuit.

The 79-part partition sums exactly: 1 + 21 + 37 + 4 + 1 + 5 + 10 = 79. It is an inventory, not a minimum-parts proposal. Exact values and individual parts: pages 15–21; editable workbook and research/cpsom_exact_components.csv.

06 / GAPS BETWEEN THE TWO REFERENCES

What needs adapting or adding?

Need	What the comparison shows	Decision for Gerald
Correct I/O voltages	CP bank supplies are all 3.3 V; old ASIC interface specifies 1.5 V and FT600 2.5 V. [C,E]	Agree bank-voltage plan, rails and any translators. No direct hookup assumed.
Data transport	Old system has FT600 + USB; CP SOM ONE has neither. [A,E]	Put transport on this PCB, carrier or another board? Which transport?
Recording hardware	ADC/front end is external to the old FPGA logic; none populated on CP. [A,B,E]	Identify exact recording/routing board, connector and timing specification.
MCU-managed boot	CP has flash and JTAG; PROGRAM_B, INIT_B and DONE are not on its three board headers. [E]	If MCU boot is chosen, route required controls/status and design recovery.
Startup defaults	PUDC_B reaches J4 pin 26 without an onboard bias in the audited netlist. [E]	Define its configuration-time state; not an unrestricted spare signal.
Recording buffer	Old main FIFO is 64 KiB on chip; CP has no external RAM. [B,E]	Confirm pause/loss requirement first; allocate Artix RAM before deciding on added memory.
Clocking	Old board uses 16 MHz input; CP author says 30 MHz. [B,C,F]	Redesign/verify clocks and constraints. Matching names is not matching timing.
Optional output/control	Old logic has stimulation controls and serial/UWB mirror.	Retain only required functions; agree safe inactive states and omitted signals.

Single-512 reference pin budget – not a final connector

22 + 2

ASIC/control + spares

+

24

FT600 bus/control

+

5

optional serial mirror

= 53 signal wires

The core ASIC interface is eight data inputs, returned clock, Read, Sync, plus eleven outputs for clocks, resets, configuration and stimulation. The two extra inputs are spare/test signals. Power, ground, JTAG and other board services are additional. [B,C]

53 being less than 76 does not prove compatibility. Clock-capable pins, I/O banks, voltages, boot functions and timing determine whether a legal pin plan exists. CP’s labels and ECP5 pin numbers are not interchangeable.

Artix bank/configuration/PCB constraints: UG471, UG470 and UG483 [G]. New regulators or translators are possibilities, not selected parts.

07 / TENTATIVE 4K TARGET

One FPGA or two? Compare both.

Assumptions for this comparison only: 4K = 4,096 simultaneous channels; eight 512-channel ASICs; equal 4+4 ASIC split if using two FPGAs; existing 16-bit words and 132/128 framing retained. Gerald must confirm these assumptions.

$$R_{\text{FPGA}} = N_{\text{per FPGA}} f_s \times 2 \times \frac{132}{128} \quad [\text{bytes/s}]$$

Allocation / metric	15,625 samples/s/ch	31,250 samples/s/ch
512-channel source baseline	16.5 MB/s	33 MB/s
One FPGA: 4,096 channels	132 MB/s per FPGA	264 MB/s per FPGA
Two FPGAs: 2,048 each	66 MB/s per FPGA	132 MB/s per FPGA
Complete 4K system, either split	132 MB/s total	264 MB/s total

Decimal MB/s; before USB/link overhead and margin. This is arithmetic from the reference format, not demonstrated FT600/MCU/UWB throughput or proof that either FPGA allocation fits.

The unchanged internal path is already too narrow for these 4K cases. At its declared 32 MHz clock, a 16-bit shared path has a $32 \times 2 = 64$ MB/s ceiling before gaps. Even the lower-rate two-FPGA case needs 66 MB/s per FPGA. A wider, faster or parallel path must be designed and verified; this is not a limit on what a redesigned Artix implementation could achieve.

The same small FIFO buys much less time

The old main FIFO is 64 KiB. If retained unchanged and initially empty, ideal time before it fills is $65,536/R$:

	Lower-rate case	Higher-rate case
One FPGA handling all 4K	0.50 ms	0.25 ms
Two FPGAs, per FPGA	0.99 ms	0.50 ms

Other buffers, occupancy and margin change actual pause tolerance. At 132 MB/s, a complete 10 ms output pause requires **1.32 MB** of total free buffer space before margin. External RAM remains a requirement-driven decision.

More channels also need more physical connections

If every 512-channel ASIC retains all **22 core interface signals independently**, eight ASICs use **176 signals** on one FPGA, or four ASICs use **88 per FPGA** in the two-way split. This excludes two optional spares per ASIC, output transport, debug, power and grounds.

CP’s existing breakout has only 76 signal wires. Either simple independent-interface scenario needs revised breakout/routing. Shared controls could reduce the count, but must be justified. The package has additional pins; 76 is the module breakout count, not the chip’s total I/O capacity.

Two FPGAs reduce per-device load, not total system data. They add a second FPGA/support circuit and require inter-device timing/channel alignment. The old four-lane framing and 512-channel host mapping also need extension. Which split fits is still open. Required decisions: actual ASIC count, shared controls/clocks, synchronization, output links, resource/timing fit and loss/pause policy. Source baseline: [A–D].

08 / ASK GERALD

Questions that the files cannot settle

Bring the prefilled answers on page 3. These questions resolve adoption, conflicts and missing system requirements; they do not ask Gerald to reconstruct the whole project.

1. **Which physical baseline are we retaining?** CP's author populated Artix-7 75T using a 35T-compatible design. What exact part/grade and board revision are in our lab? Which FPGA_512 tree/bitstream is the working reference?
2. **Does 4K mean 4,096 simultaneous channels from eight 512-channel ASICs?** One FPGA or two, and what determines the split? Which controls/clocks can be shared, and how will devices stay synchronized?
3. **Which rate is required: 15,625, 31,250 or another rate?** What are the measured outgoing and returned clocks? The slides support 31,250; the host uses 15,625. Can we have the timing specification and explanation for unused Sync?
4. **What is the final data route?** Retain FT600 USB, use a different bridge/MCU, or retain the serial/UWB output? For two FPGAs: independent output links or one aggregate link, where do data merge and how are channel identity/timing aligned? Which optional functions stay?
5. **Which MCU and routing board are used?** What is their exact role, pinout, connector and signal voltage? Are the old 1.5 V / 2.5 V levels still applicable?
6. **What must happen when the computer pauses?** Maximum stall, loss policy, overflow response, recording duration and file-size handling? How are missing samples marked?
7. **Who supplies power, clock and boot configuration?** Keep local flash or use MCU boot? What startup/shutdown/recovery behavior and current/heat budget are required?
8. **What does "small enough" mean?** Module or complete assembly; length, width, height, mounting, connector placement, layers and assembly limits?

Documents to request

Exact board/ASIC revisions and photos; recording/routing schematic and pinout; known-good bitstream and host version; ASIC configuration map/settings; clock timing; mechanical envelope and supply/current measurements.

Record the most important answers

Decision / source / owner:

Still unresolved / owner / due date:

No voltage measurement, exact new-board rate, MCU selection, loss policy, current budget or dimension target is supplied by this comparison. Keep unresolved items open.

09 / CHANGE WORKSHEET

Start with the board, then mark changes

K = keep, C = change, R = remove, M = move elsewhere, A = add. Mark only after agreeing the condition on pages 5–6. A suggestion is not an approved deletion.

Block / feature	Decision	Check / owner / due
Chosen FPGA U1	KEEP – agreed	Record exact ordering code.
Connectors + mounting	☐ K ☐ C ☐ R	TBD
Power generation (21 parts)	☐ K ☐ C ☐ M	TBD
Local rail capacitors (37)	☐ K ☐ C	TBD; validated network.
Flash + pulls (4)	☐ K ☐ C ☐ M/R	TBD; boot replacement.
Oscillator (1)	☐ K ☐ C ☐ M/R	TBD; valid clock source.
Startup resistors + boot mode	☐ K ☐ C	TBD; defined states.
LEDs + resistors (10)	☐ K ☐ C ☐ R	TBD; review DONE behavior.
I/O voltage adaptation	☐ C ☐ A	TBD; bank/power plan.
Data transport / USB function	☐ A on PCB ☐ M to carrier	TBD; interface choice.
MCU boot/control connections	☐ A ☐ Not needed	TBD; MCU role first.
External recording RAM	☐ A ☐ Not needed	TBD; buffer budget first.

Next deliverable: agreed 4K definition and one/two-FPGA split, then a voltage/pin map and revised block diagram. Then compare component placements and routed layouts. Do not approve a minimum part count from this worksheet alone.

Evidence for the revised design: FPGA implementation/timing; reviewed power and pin plan; PCB checks; measured boot, maximum-rate transfer, buffer behavior, temperature and safe power transitions. Editable individual-part inventory: Gerald_Component_Review.xlsx and research/cpsom_exact_components.csv. Unselected choices remain open.

10 / SOURCES AND TERMS

What each statement is based on

FPGA_512 snapshot: [commit 767e825](#), copied into reference/FPGA_512.

CP SOM ONE snapshot: [commit b272899](#), in reference/cp_som_one. Both reviewed 16 September 2026.

ID	Primary source / what it establishes
[A]	FPGA_512 README; adc_4lane_framed/adc_live.py and adc_daemon.c: channel/rate assumptions, host transport, recording and error behavior.
[B]	adc_4lane_framed/top_4lane_framed.v; fpga/pair_tdm_framer.v and async_fifo.v: ports, sample/frame format, clocks and FIFO size.
[C]	adc_4lane_framed/adc_4lane_framed.lpf: ECP5 pin/electrical intent and clock constraints. Values are not measurements.
[D]	adc_4lane_framed/Makefile and FPGA board.net: ECP5 build/programming and old hardware components.
[E]	CP kicad/som_fpga.xml, som_fpga.kicad_pcb, supply.kicad_sch and production_files/som_fpga.csv: circuit, layout and inventory.
[F]	Author's CP SOM ONE design article : 35T/75T rationale, intended 30 MHz oscillator and design context.
[G]	AMD DS181, UG470, UG471, UG475 and UG483: electrical limits, configuration, I/O banks, package/pinout and PCB support. Local PDFs in Datasheets/AMD.
[H]	Gerald's supplied <i>Chip_FPGA Interface.pptx</i> , 28 slides. Source deck, matching PDF and per-slide evidence saved under research/gerald_slides.

Detailed evidence and editable inventory: [START_HERE.md](#) links to both audit notes and the component checklist. The notes contain exact file/line pointers, calculations and open requirements.

Terms: **RTL** = code describing digital hardware. **FIFO** = first-in, first-out data buffer. **CRC** = corruption check. **I/O bank** = group of FPGA pins sharing voltage rules. **FT600** = USB bridge, not the proposed MCU. **Flash** stores boot data; FPGA block RAM buffers working data.

Print this packet: 26 pages / 13 sheets double-sided. The existing complete binder contains the older guides and full chip manuals; use this packet for current scope. Source comparison does not prove hardware compatibility or successful Artix implementation.

Image license

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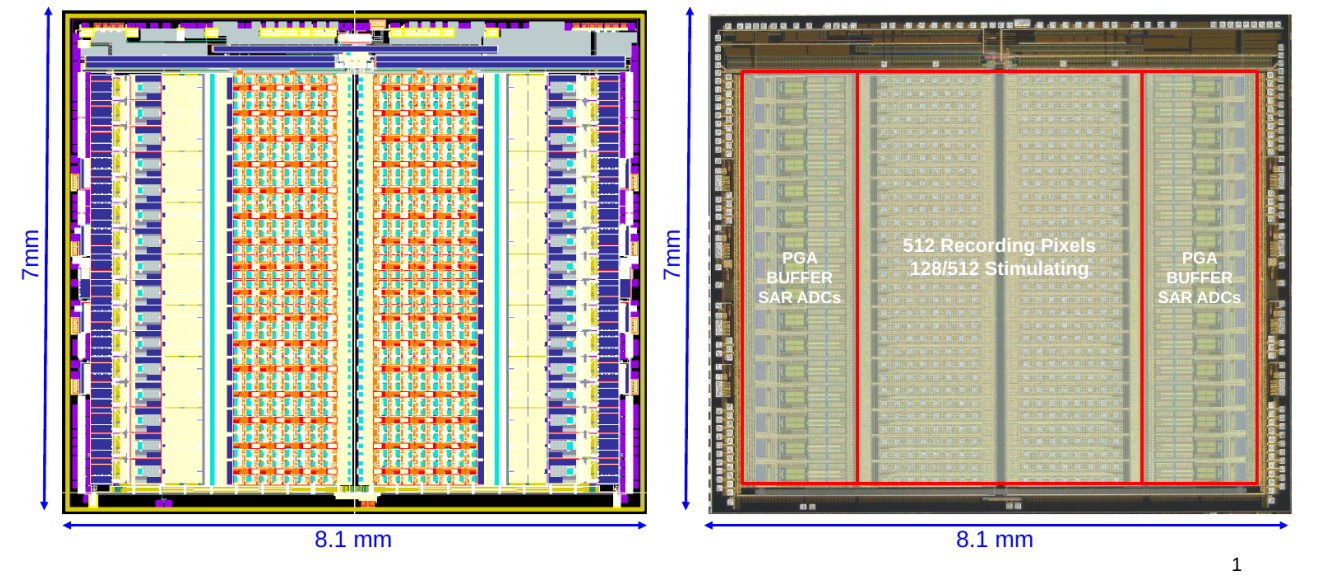
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The ASIC is the recording front end

ASIC Layout

RISE



Gerald's supplied presentation, slide 1. Reproduced for this internal meeting packet. [H]

The slides explain the device that FPGA_512 talks to. The ASIC contains the recording amplifiers and ADCs. The FPGA receives already digitized data, organizes it and sends it to the computer. Shrinking the FPGA PCB does not eliminate the ASIC's clock, reset, configuration or data connections.

Slide evidence	Connection to the old code	PCB consequence
512 recording channels; 32 modules of 16 amplifiers. Eight data wires, four ADCs per wire. Slides 1, 5, 14.	Matches 8 physical data inputs, 64 channels per input, then four paired streams.	One ASIC does not need 512 FPGA input pins. It needs serial data and control/timing wires.
Chip reset, front-end reset, outgoing clock. Returned clock, Read and Sync. Slide 14.	Correspond to chip clock/reset, capture clock, Read and Sync ports.	Reserve these connections and suitable clock pins. Resolve the unused Sync input in the old RTL.
Configuration and stimulation interfaces. Slides 25, 27.	Configuration CLK/L/R/LATCH and four stimulation outputs appear in the RTL.	These are ASIC controls, distinct from FPGA boot flash and JTAG. Keep them if the required ASIC operation needs them.

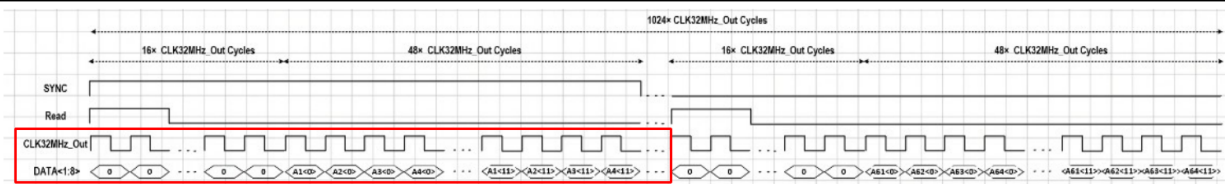
SPI also programs amplifier gain. Recording-only operation still needs this configuration path unless another validated method replaces it. Stimulation's four control outputs are a separate decision. ASIC revision, connector and digital timing/voltage specification remain open.

12 / RATE AND CLOCK EVIDENCE

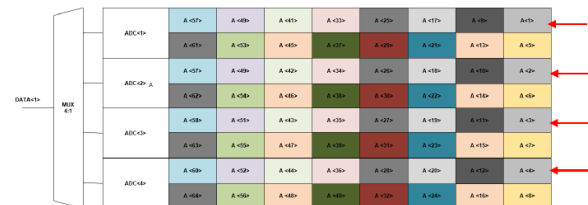
The slides narrow the timing question

Communication Protocol

RISE



- **1 Frame = 1024 clock cycles**
- **Within each frame, there is one Sync and 16 Read**
- **Sync, signals when the FPGA should start streaming the data → aligns the channel map**
- **Read signals a new amplifier is being streamed**
 - Within each Read period, DATA streams information about 4 amplifiers (from 4 different ADCs which share the same data line)
 - They all have the same location in a group of 16



Gerald's slide 15. The waveform is source material, not a timing measurement made in this audit. [H]

$$f_s = \frac{f_{\text{returned clock}}}{1024}, \quad \frac{32 \text{ MHz}}{1024} = 31,250 \text{ samples/s/channel.}$$

Slide 15 specifies a 1,024-clock frame with 16 Read periods. Each Read period carries four 12-bit samples per data wire. Slides 3 and 6 explicitly state **31.25 kS/s/channel**. This supports the higher-rate design scenario on page 7.

The clock sources disagree. Slide 13 names a 32 MHz FPGA-to-ASIC clock. Current RTL forwards 16 MHz; its constraint file still specifies a 32 MHz returned clock. The host uses 15,625 samples/s/channel. A returned 16 MHz clock with 1,024 cycles/frame would explain that host rate, but that relationship has not been measured or established.

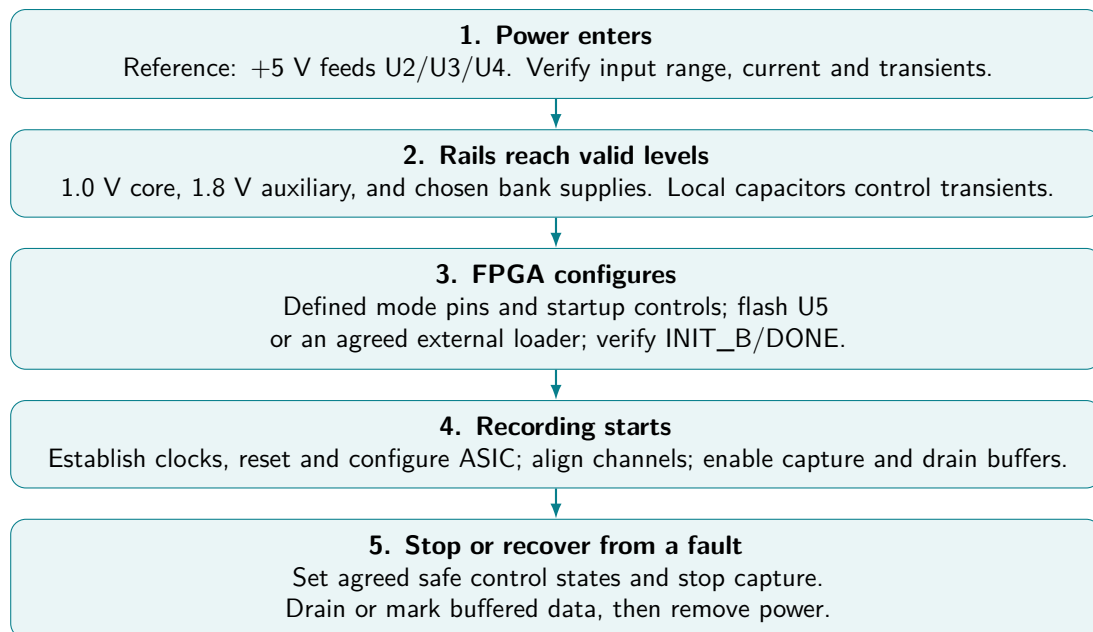
Three questions to settle with Gerald

1. Are we building for 31.25 kS/s/channel, 15.625 kS/s/channel, or selectable modes? Which bitstream and ASIC revision implement that mode?
2. What are the actual outgoing and returned clock frequencies and phase/edge relationships? Please confirm with the ASIC timing specification or a scope capture.
3. Slide 15 uses Sync for channel alignment, but the old top-level Sync input is unused. How does the working system establish and recover the correct channel map?

If stimulation remains: its timing also needs reconciliation. Slide 18 says 64 cycles = 2 ms; slides 27–28 label 60 kHz; current RTL implies a 16 kHz square wave at a 32 MHz input. These values do not agree.

Full 28-slide deck and extracted evidence: [research/gerald_slides](#). Exact slide/code links: [research/gerald_slides_audit.md](#).

13 / POWER ON TO POWER OFF

What the smaller board must still do

Observed enable wiring: U3 and U4 EN connect to +5 V; U2 EN connects to +1.8 V. This is not a measured, supervised sequence. Verify ramps, reset release and brownout behavior against DS181/UG470 for the retained part. [E,G] **Recording controls are a separate step from booting the FPGA.** Loading the bitstream does not establish ASIC gain, resets, channel alignment or stimulation state. Gerald's slides describe these runtime controls. [H]

Ask Gerald: who controls power and resets, what happens if USB/MCU/ASIC powers up first, and what should happen on cable loss or low voltage? Define unpowered-pin behavior to prevent unintended powering through I/O. Preserve a usable programming/recovery connection on the compact board.

This is a proposed system review sequence. Actual sequencing, discharge timing, reset durations and stimulation-safe states remain requirements to agree, not validated behavior of CP SOM ONE.

14 / HOW TO USE THE COMPONENT AUDIT

Required function versus replaceable part

A component’s reference value is not a universal minimum. For example, C7 is a specific 4.7 μ F capacitor in this reference. The requirement is a power network that keeps the FPGA rail within limits under the final load. Whether C7’s value, package or placement changes follows from that design and its validation.

Constraint	What is fixed / required	What may change
Exact FPGA retained	Its 23 $\times$ 23 mm FGG484 body, ball pitch and electrical limits. Resolve full ordering code.	PCB outline, placement, stack-up and routes around it. No smaller package in this task.
Power integrity	Valid rails and adequate local transient supply at FPGA pins.	Regulator choice/location; capacitor value, quantity, package and placement after analysis.
Configuration	Reliable configuration on each startup, defined mode/control pins and recovery.	Onboard flash versus supported external loading; connector versus test pads.
Clock and interfaces	Required clocks, legal voltage levels, correct timing and enough connections.	Clock source, bank allocation, connector form and control sharing if justified.
Mechanical assembly	BGA escape, fabrication/assembly limits, thermal path and attachment.	Margins, mounting arrangement and board outline once requirements are known.

How to read the next pages

Reference identifies one placed part. **Value** is transcribed from source. **Package** describes the body/land pattern. **Connection** is traced from the exported netlist and checked against the PCB. **Review** is a proposed action, not Gerald’s approval. **Package shorthand:** 0402 = nominal 1.0 $\times$ 0.5 mm body; 0603 = 1.6 $\times$ 0.8 mm; 0805 = 2.0 $\times$ 1.25 mm; 1206 = 3.2 $\times$ 1.6 mm. Actual land pattern and clearances are larger. Exact footprints and supplier codes remain in the workbook.

The repository alone is not a fully specified purchasing BOM. Its passive fields omit many ratings and manufacturer part numbers. Additional catalog research now resolves all five capacitor identities and ratings (page 23). Actual fitted parts and effective capacitance remain unconfirmed; resistor ratings and other unresolved ordering details remain open.

Audit scope: all 79 electronic placements, plus four connectors, four mounting holes and the U6 logo (88 source objects total). The logo is not a sixth IC. Original files remain unchanged. Source snapshot [E]. Proposed reviews also use [G] and the TI TLV62565, onsemi NCP1529, Macronix MX25L12833F and Epson SG-8018 documents in Datasheets.

15 / EXACT CAPACITORS

Input, output and 3.3 V capacitors

One row per component. All listed capacitor pin 2 connections are GND. Values are nominal source values; effective capacitance depends on the selected part and operating voltage. K/C/M/R choices remain open.

Ref.	Value	Pkg.	Connection / role	Side	BOM code	Review
C1	4.7 μ F	0805	+5 V; U2 input	Top	C162420	Q1 _____
C2	10 μ F	0805	+3.3 V; U2 output	Top	C105505	Q1 _____
C3	4.7 μ F	0805	+5 V; U3 input	Top	C162420	Q1 _____
C4	10 μ F	0805	+1.8 V; U3 output	Top	C105505	Q1 _____
C5	4.7 μ F	0805	+5 V; U4 input	Top	C162420	Q1 _____
C6	10 μ F	0805	+1.0 V; U4 output	Top	C105505	Q1 _____
C7	4.7 μ F	0805	+3.3 V; rail capacitor	Bot.	C162420	Q3 _____
C8	4.7 μ F	0805	+3.3 V; rail capacitor	Bot.	C162420	Q3 _____
C9	4.7 μ F	0805	+3.3 V; rail capacitor	Bot.	C162420	Q3 _____
C10	4.7 μ F	0805	+3.3 V; rail capacitor	Top	C162420	Q3 _____
C11	4.7 μ F	0805	+3.3 V; rail capacitor	Bot.	C162420	Q3 _____
C12	4.7 μ F	0805	+3.3 V; rail capacitor	Bot.	C162420	Q3 _____
C13	4.7 μ F	0805	+3.3 V; rail capacitor	Bot.	C162420	Q3 _____
C14	0.47 μ F	0402	+3.3 V; rail capacitor	Bot.	C106255	Q3 _____
C15	0.47 μ F	0402	+3.3 V; rail capacitor	Top	C106255	Q3 _____
C16	0.47 μ F	0402	+3.3 V; rail capacitor	Bot.	C106255	Q3 _____

Q1: Will the regulator circuit stay here?

Keep input/output filtering with its regulator unless a redesigned supply replaces that function. Gerald: confirm the actual regulator, required current, input source and stable capacitor/inductor network. Moving generation does not remove the FPGA's need for local decoupling.

Q3: What does the final power network require?

Keep these rail capacitors as the starting network. Gerald: approve the load estimate and rail/bank plan, then check manufacturer guidance, DC-bias loss, impedance and placement before changing value, quantity or package.

[E] per-reference BOM, schematic, XML and PCB audit. Exact source line pointers and native PCB X/Y coordinates are in the workbook/CSV. Q1–Q3 are review questions, not approvals.

16 / EXACT CAPACITORS

3.3 V and 1.8 V rail capacitors

One row per component. All listed capacitor pin 2 connections are GND. Values are nominal source values; effective capacitance depends on the selected part and operating voltage. K/C/M/R choices remain open.

Ref.	Value	Pkg.	Connection / role	Side	BOM code	Review
C17	0.47 μ F	0402	+3.3 V; rail capacitor	Top	C106255	Q3 _____
C18	0.47 μ F	0402	+3.3 V; rail capacitor	Top	C106255	Q3 _____
C19	0.47 μ F	0402	+3.3 V; rail capacitor	Bot.	C106255	Q3 _____
C20	0.47 μ F	0402	+3.3 V; rail capacitor	Bot.	C106255	Q3 _____
C21	0.47 μ F	0402	+3.3 V; rail capacitor	Bot.	C106255	Q3 _____
C22	0.47 μ F	0402	+3.3 V; rail capacitor	Bot.	C106255	Q3 _____
C23	47 μ F	1206	+3.3 V; rail capacitor	Top	C385907	Q3 _____
C24	47 μ F	1206	+1.8 V; rail capacitor	Top	C385907	Q3 _____
C25	4.7 μ F	0805	+1.8 V; rail capacitor	Bot.	C162420	Q3 _____
C26	4.7 μ F	0805	+1.8 V; rail capacitor	Bot.	C162420	Q3 _____
C27	4.7 μ F	0805	+1.8 V; rail capacitor	Bot.	C162420	Q3 _____
C28	4.7 μ F	0805	+1.8 V; rail capacitor	Bot.	C162420	Q3 _____
C29	0.47 μ F	0402	+1.8 V; rail capacitor	Bot.	C106255	Q3 _____
C30	0.47 μ F	0402	+1.8 V; rail capacitor	Bot.	C106255	Q3 _____
C31	0.47 μ F	0402	+1.8 V; rail capacitor	Bot.	C106255	Q3 _____
C32	0.47 μ F	0402	+1.8 V; rail capacitor	Bot.	C106255	Q3 _____

Q3 applies to every row

The 3.3 V rail feeds the shown I/O/configuration banks, flash and oscillator. The 1.8 V rail feeds auxiliary, ADC and battery-supply pins in this reference. Changing bank voltages changes the required rail partition and capacitor assignment.

A shared net does not make every placement redundant. Nearby small capacitors and bulk capacitance serve different frequency/current paths. Count and nominal microfarads alone cannot establish which capacitor can be deleted.

[E] per-reference BOM, schematic, XML and PCB audit. Exact source line pointers and native PCB X/Y coordinates are in the workbook/CSV. Q1–Q3 are review questions, not approvals.

17 / EXACT CAPACITORS

1.0 V capacitors and feedback parts

One row per component. All listed capacitor pin 2 connections are GND. Values are nominal source values; effective capacitance depends on the selected part and operating voltage. K/C/M/R choices remain open.

Ref.	Value	Pkg.	Connection / role	Side	BOM code	Review
C33	100 μ F	1206	+1.0 V; rail capacitor	Top	C268006	Q3 _____
C34	100 μ F	1206	+1.0 V; rail capacitor	Bot.	C268006	Q3 _____
C35	4.7 μ F	0805	+1.0 V; rail capacitor	Bot.	C162420	Q3 _____
C36	4.7 μ F	0805	+1.0 V; rail capacitor	Bot.	C162420	Q3 _____
C37	4.7 μ F	0805	+1.0 V; rail capacitor	Bot.	C162420	Q3 _____
C38	4.7 μ F	0805	+1.0 V; rail capacitor	Bot.	C162420	Q3 _____
C39	4.7 μ F	0805	+1.0 V; rail capacitor	Bot.	C162420	Q3 _____
C40	0.47 μ F	0402	+1.0 V; rail capacitor	Bot.	C106255	Q3 _____
C41	0.47 μ F	0402	+1.0 V; rail capacitor	Bot.	C106255	Q3 _____
C42	0.47 μ F	0402	+1.0 V; rail capacitor	Bot.	C106255	Q3 _____
C43	0.47 μ F	0402	+1.0 V; rail capacitor	Bot.	C106255	Q3 _____
C44	0.47 μ F	0402	U2 FB to GND	Top	C106255	Q2 _____
C45	0.47 μ F	0402	U3 FB to GND	Top	C106255	Q2 _____
C46	0.47 μ F	0402	U4 FB to GND	Top	C106255	Q2 _____

Q2: Why are C44, C45 and C46 on feedback?

Each is 0.47 μ F from a regulator FB pin to ground. This differs from the TI illustrated circuit and the onsemi feed-forward capacitor connection across the upper divider. Gerald: confirm the populated regulator and intended compensation, then review stability/startup before retaining or changing them.

Exact count check: 19 of 4.7 μ F; 3 of 10 μ F; 20 of 0.47 μ F; 2 of 47 μ F; 2 of 100 μ F: **46 capacitors total**. Packages: 22 of 0805, 20 of 0402 and 4 of 1206.

[E] per-reference BOM, schematic, XML and PCB audit. Exact source line pointers and native PCB X/Y coordinates are in the workbook/CSV. Q1–Q3 are review questions, not approvals.

18 / EXACT RESISTORS

Power dividers and startup settings

All 19 resistors use 0402 footprints. This page lists R1–R11. The connection wording below expands source net names; the workbook preserves exact pin-to-net strings.

Ref.	Value	Exact function / connection	Side	BOM code	Review
R1	100 kΩ	3.3 V to U2 FB; upper divider	Top	C131362	Q1 _____
R2	22 kΩ	U2 FB to GND; lower divider	Top	C409681	Q1 _____
R3	120 kΩ	U3 FB to GND; lower divider	Top	C323683	Q1 _____
R4	240 kΩ	1.8 V to U3 FB; upper divider	Top	C323663	Q1 _____
R5	22 kΩ	1.0 V to U4 FB; upper divider	Top	C409681	Q1 _____
R6	33 kΩ	U4 FB to GND; lower divider	Top	C416388	Q1 _____
R7	4.7 kΩ	CFGBVS to 3.3 V; configuration voltage	Bot.	C3017492	Q4 _____
R8	4.7 kΩ	M2/J1 pin 2 to GND; boot mode	Top	C3017492	Q4 _____
R9	4.7 kΩ	INIT_B to 3.3 V; startup/error status	Bot.	C3017492	Q4 _____
R10	4.7 kΩ	PROGRAM_B to 3.3 V; configuration reset	Bot.	C3017492	Q4 _____
R11	330 Ω	DONE to 3.3 V; external pull-up	Top	C881192	Q4 _____

The divider pairs explain the nominal rails

$$V_{\text{out}} = V_{\text{FB}} \left(1 + \frac{R_{\text{upper}}}{R_{\text{lower}}} \right)$$

At a nominal 0.6 V feedback reference: U2 uses R1/R2 (100k/22k), giving 3.327 V; U3 uses R4/R3 (240k/120k), giving 1.8 V; U4 uses R5/R6 (22k/33k), giving 1.0 V. These calculations do not validate current capacity, tolerances or loop stability.

Q4: review R8 before copying. The source uses 4.7 kΩ for M2. UG470 v1.17 p.20 specifies a direct connection or ≤1 kΩ for mode pins. Ask Gerald to approve the corrected boot-mode circuit, including J1 and recovery. R9/R10 provide external pull functions described on p.21. R11 is separate from the DONE LED resistor R18; external 330 Ω DONE pull-up is optional under the conditions in p.22.

[E,G]; TI TLV62565 feedback reference and application section; onsemi NCP1529 archived Rev.5 pp.13–15. R7 must agree with the final configuration-bank voltage. No resistor tolerance or power rating is explicit in the repository BOM value.

19 / EXACT RESISTORS AND LEDS

Flash pulls and removable indicators

Ref.	Value	Connection / role	Side	BOM code	Review
R12	10 k Ω	USER_LED1 to D3 anode	Top	C328298	Q6
R13	10 k Ω	+5 V to D2 anode; input-power LED	Top	C328298	Q6
R14	10 k Ω	USER_LED2 to D4 anode	Top	C328298	Q6
R15	4.7 k Ω	+3.3 V to flash CS	Bot.	C3017492	Q5
R16	4.7 k Ω	+3.3 V to flash D2	Bot.	C3017492	Q5
R17	4.7 k Ω	+3.3 V to flash D3	Bot.	C3017492	Q5
R18	100 Ω	DONE to D1 anode; LED current resistor	Top	C471762	Q6
R19	10 k Ω	USER_LED0 to D5 anode	Top	C328298	Q6

All five LED cathodes connect to ground

Ref.	Source label	Package / matching resistor	Side	BOM code	Review
D1	DONE	0603; paired with R18	Top	C89814	Q6
D2	VCC	0603; paired with R13	Top	C601673	Q6
D3	USER_LED	0603; paired with R12	Top	C40378	Q6
D4	USER_LED	0603; paired with R14	Top	C40378	Q6
D5	USER_LED	0603; paired with R19	Top	C40378	Q6

Q5: Will we retain local SPI boot flash?

R15–R17 keep flash control/data pins biased. Keep/review them with U5 if using local SPI boot. Removing flash requires a supported configuration source, power-up handshake and recovery design; the future MCU is not selected yet.

Q6: Which indicators are worth their area?

D2/R13 and D3/R12, D4/R14, D5/R19 are removal candidates if their status/debug functions can be provided elsewhere. Review D1/R18 together with the complete DONE circuit. **Removing the DONE LED does not automatically mean removing R11.**

Proposed meeting choice:

☐ Keep all ☐ Keep selected ☐ Remove indicators

Selected pairs / reason: _____

[E]. LED color, forward voltage/current and exact manufacturer ordering code are not established by the source value labels DONE/VCC/USER_LED. Supplier codes are preserved for purchasing verification.

FPGA, regulators, flash, clock and inductors

Ref.	Source identity, connection and proposed review	BOM code	Side
U1	BOM: XC7A75T-FGG484. Schematic/PCB: XC7A35T-FGG484. 23 × 23 mm, 484 balls. Keep chosen device; resolve full ordering code.	C396343	Top
U2	TLV62565DBVR value, SOT-23-5. +5 V to +3.3 V; EN on +1.8 V. Symbol metadata is NCP1529A. Q1.	C87122	Top
U3	TLV62565DBVR value, SOT-23-5. +5 V to +1.8 V; EN on +5 V. Supplier code points to NCP1529ASNT1G. Q1.	C23902	Top
U4	TLV62565DBVR value, SOT-23-5. +5 V to +1.0 V; EN on +5 V. Supplier code points to NCP1529ASNT1G. Q1.	C23902	Top
U5	MX25L12833F, 128 Mbit (16 MiB), WSON 6 × 5 mm. Metadata links another flash family/capacity. Q5.	C2802709	Top
X1	SG-8018CE; 3.2 × 2.5 mm; 3.3 V VCC/OE. OUT to FPGA C18 (MRCC bank 16). Article says 30 MHz; full suffix unspecified. Q7.	C390488	Top
L1	2.2 μH; U2 SW to +3.3 V. Q1.	C341775	Top
L2	2.2 μH; U3 SW to +1.8 V. Q1.	C341775	Top
L3	2.2 μH; U4 SW to +1.0 V. Q1.	C341775	Top

L1–L3 footprint: inductors:IND_LQH3NPN2R2MJRL. Verify actual inductor ordering code, saturation/RMS current, DCR and regulator requirements before shrinking. A footprint name alone does not identify the fitted part.

Q7: What clock does the recording implementation need?

Confirm X1’s full frequency/voltage/enable specification and the Artix clock plan. The 30 MHz reference oscillator and old nominal 16 MHz implementation are different starting points. A smaller/shared clock source is conditional on jitter, routing, startup and timing requirements.

75T routing opportunity, not a completed pin plan: the 35T symbol leaves 35 package pads with no net; official 35T pinout marks them NC, while the 75T pinout identifies bank-13 I/O. This helps explain unused package capacity. Rebuild and validate the pin plan for the exact retained device; do not assume all can serve the required clock/voltage interfaces.

[E,F,G]. All nine parts are included in the 79-placement assembly BOM. The actual assembled models and oscillator suffix still need confirmation.

21 / CONNECTORS AND MECHANICS

All nine objects outside the assembly BOM

Ref.	Exact source object	Connection / proposed review
J1	2-pin 2.54 mm boot selector; top.	Pin 1: 3.3 V. Pin 2: M2 with R8 pull-down. DNP flag in source. Q4: fixed strap/pads or connector?
J2	40-pin Samtec TFM footprint; bottom.	CONN3: 20 signal wires plus JTAG. Q8: replace with the agreed recording/service connector?
J3	40-pin Samtec TFM footprint; bottom.	CONN1: 28 signal wires. Q8: size after complete pin/voltage budget.
J4	40-pin Samtec TFM footprint; bottom.	CONN2: 28 signal wires; pin 26 includes PUDC_B. Q8: define startup state.
H1	3.2 mm M3 mounting hole/pad.	GND-connected. Q8: retain, move or change attachment after mechanical agreement.
H2	3.2 mm M3 mounting hole/pad.	GND-connected. Q8: retain, move or change attachment after mechanical agreement.
H3	3.2 mm M3 mounting hole/pad.	GND-connected. Q8: retain, move or change attachment after mechanical agreement.
H4	3.2 mm M3 mounting hole/pad.	GND-connected. Q8: retain, move or change attachment after mechanical agreement.
U6	cp_logo25mm; graphic footprint.	No electrical pads. May resize/remove artwork; preserve applicable source-license notices.

J2–J4 footprint: samtec:SAMTEC_TFM-120-02-S-D-A. Pitch is 1.27 mm. These connector footprints are present but have no production-BOM supplier code. The complete mating connector, stacking height and assembly selection still need definition. **Common power contacts:** J2, J3, J4 pins 1/2 = +5 V; pins 39/40 = +3.3 V. **JTAG on J2:** 5 TMS, 7 TCK, 9 TDO, 11 TDI; grounds 6/8/10 and 3.3 V reference on 12. Full per-pin map is in the audit JSON and workbook source details.

Q8: define interfaces before shrinking connectors. At about 4K channels, connector signal count, ground returns and voltage domains may require more routing even if some general-purpose headers disappear. PUDC_B has no onboard bias in this source; UG470 p.22 requires a defined level.

Count reconciliation: 46 capacitors + 19 resistors + 5 LEDs + 3 inductors + U1–U5 + X1 = 79. Add J1–J4, H1–H4 and U6 = 88. All 88 references were found in the schematics, XML and PCB. All 748 XML pin/net entries matched PCB pad-net strings; this is not an ERC/DRC or routed-performance test. [E]. H1–H4 and U6 also carry DNP flags; this flag is not evidence that the footprint/hole is absent from the board. The 79 assembly placements split into 43 top and 36 bottom.

22 / COMPONENT DECISIONS WITH GERALD

Eight questions tied to exact references

Record a decision, evidence and owner. The exact per-part workbook has editable decision fields; all unagreed choices remain TBD.

ID	Parts affected	Question / meeting record
Q1	U2–U4, L1–L3, R1–R6, C1–C6	Which regulator models are actually fitted? Keep, change or move each supply? What measured/estimated current and capacitor network supports the final recording load? Decision/owner: _____
Q2	C44–C46	Why is 0.47 microfarad connected from each FB pin to ground? Is this the intended and validated loop circuit for the selected regulators? Decision/owner: _____
Q3	C7–C43	What rail and bank allocation does our exact FPGA need? Which capacitor value/package/placement changes can we justify after load and power-network review? Decision/owner: _____
Q4	R7–R11, J1; PUDC_B	Approve boot mode and biasing: address R8 versus the UG470 mode-pin limit, retain valid INIT/PROGRAM behavior and define PUDC_B and DONE handling. Decision/owner: _____
Q5	U5, R15–R17	Keep onboard flash or specify an external loader? If MCU boot, which MCU, boot mode, control/status pins and recovery process? Decision/owner: _____
Q6	D1–D5, R12–R14, R18–R19	Which indicator pairs may be removed? How will we diagnose power and configuration without them? Review R11 separately from D1/R18. Decision/owner: _____
Q7	X1	Which outgoing/returned frequencies and sample rate are required? Which oscillator specification and Artix clock implementation meet timing? Decision/owner: _____
Q8	J2–J4, H1–H4	What total assembly dimensions and connector/pin/bank/ground plan support one or two FPGAs at the agreed channel count? Decision/owner: _____

Specific documents behind the electrical questions

TI TLV62565 Rev.D, application pp.10–13; onsemi NCP1529 archived Rev.5, pp.13–15; AMD UG470 v1.17, pp.20–22; AMD UG483, decoupling guidance. See `research/component_design_review.md` for exact comparisons and source links. These manufacturer checks constrain changes; they do not establish a final minimum BOM. **Meeting outcome to aim for:** agree the requirements and mark each affected part K/C/R/M (keep/change/remove/move), or leave it TBD with an owner and the measurement/document needed.

23 / CAPACITOR PURCHASING EVIDENCE

Exact identities behind all 46 capacitors

These are verified supplier-code mappings, not guesses from the footprint. They identify what the public BOM codes currently point to. They do not prove what is fitted on the lab board or approve a replacement.

BOM code	Manufacturer / exact catalog part	Value	Rating	Tol.	Case	Qty.
C162420	Murata GRM21BR61H475KE51L	4.7 μ F	50 V	$\pm 10\%$	0805	19
C105505	Walsin 0805X106K100CT	10 μ F	10 V	$\pm 10\%$	0805	3
C106255	YAGEO CC0402KRX5R6BB474	0.47 μ F	10 V	$\pm 10\%$	0402	20
C385907	Taiyo Yuden EMK316BBJ476ML-T	47 μ F	16 V	$\pm 20\%$	1206	2
C268006	Taiyo Yuden JMK316ABJ107ML-T	100 μ F	6.3 V	$\pm 20\%$	1206	2

All five use **X5R dielectric**. C106255's 470 nF is the same as 0.47 μ F. Quantity is placement count in the reference BOM, not an order quantity. All 46 capacitor references are individually mapped on pages 15–17 and in the workbook.

A concrete size-reduction question

The 4.7 μ F source choice is a **50 V, 0805** capacitor, even where its circuit rail is only 1.0, 1.8, 3.3 or 5 V. That creates a useful question for Gerald: can a smaller case with an appropriate voltage rating provide the required *effective* capacitance, impedance and transient performance? A lower voltage label alone is not enough to select it.

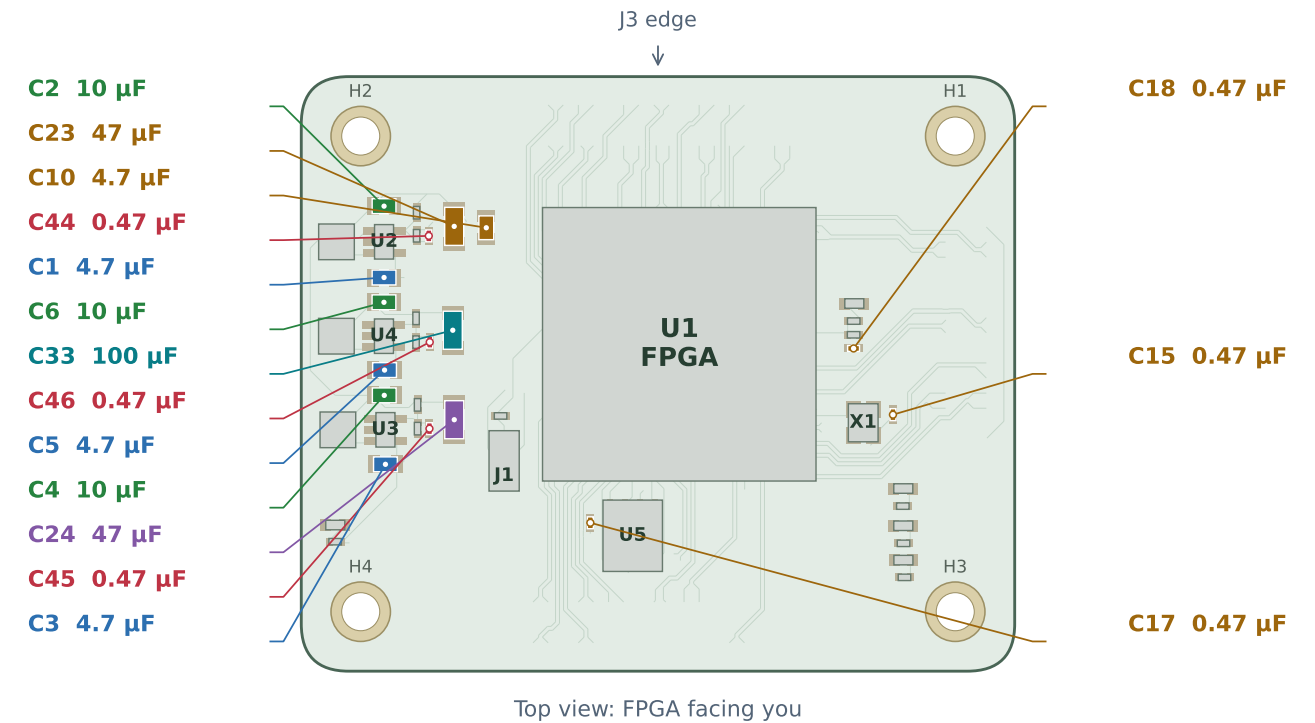
How to decide if a smaller capacitor is acceptable: identify its rail and job; set required effective capacitance and impedance; check voltage transients, tolerance, DC-bias/temperature loss and ESR/ESL; check placement and assembly; validate the regulator/FPGA power behavior. The same nominal microfarads in a smaller package do not guarantee the same behavior.

Evidence and remaining gaps

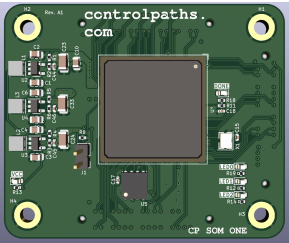
Catalog identities and properties were checked on 16 September 2026. Manufacturer evidence corroborates Murata, YAGEO and both Taiyo Yuden entries. Walsin's catalog mapping and a linked family approval sheet are available; an exact-part DC-bias curve remains unverified. Taiyo Yuden's current site maps the older names to newer names; that mapping is not an approved substitution. **Meeting question:** which physical board/BOM confirms the actual fitted parts, and who will evaluate smaller candidate parts against the final power requirements? Primary supplier identity links are clickable in the first column. Manufacturer links, exact references, scope and verification status are in `research/capacitor_catalog_mapping.json` and the workbook's Capacitor specs sheet.

Top side: 16 labeled capacitors

Follow a colored line to the white dot on the part. C1, C2, etc. are the original PCB reference numbers. The value beside each number tells you how much capacitance it has.



Color / function	Which parts?	What they do
Input support	C1, C3, C5	Supply short bursts of current to each switching regulator from the 5 V input.
Output filtering	C2, C4, C6	Work with L1/L2/L3 to smooth the regulator outputs and support load changes.
3.3 V support	C10, C15, C17, C18, C23	Help keep the shared 3.3 V rail steady for I/O, configuration, flash and clock circuitry.
1.8 V support	C24	Larger local charge reserve for the auxiliary supply network.
1.0 V support	C33	Larger local charge reserve for FPGA core and block-RAM power.
Feedback network	C44, C45, C46	Connect to regulator voltage-sensing pins. Their intended effect and stability need review.

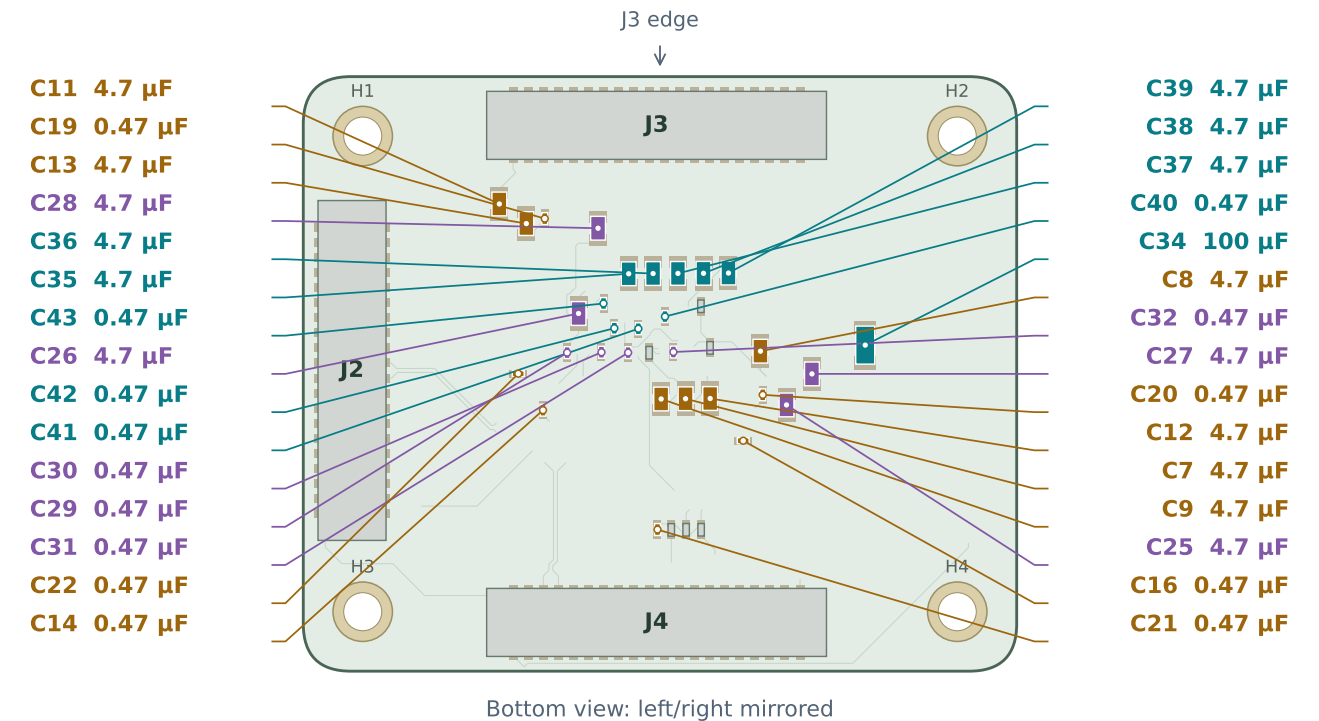


Match it to the original picture. The FPGA is in the middle; the three regulators are on the left. From top to bottom they are **U2 (3.3 V)**, **U4 (1.0 V)**, **U3 (1.8 V)**. Their physical order is different from their reference-number order. Left: original GitHub render, ControlPaths [E]. Above: vector drawing from the same KiCad PCB coordinates and footprints; copper is simplified for readability. Values/nets: [E].

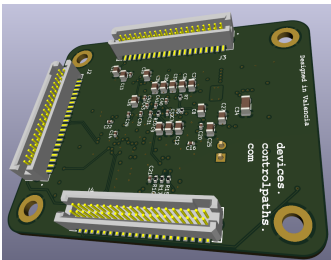
25 / TURN THE BOARD OVER

Bottom side: the other 30 capacitors

Look directly at the underside, keeping the J3 edge at the top. Left and right are mirrored relative to the top view. J2 is now on the left. Each label points to the exact reference footprint.



Rail / count	Capacitors on this side	Function
3.3 V / 12	4.7 μ F: C7, C8, C9, C11, C12, C13. 0.47 μ F: C14, C16, C19, C20, C21, C22.	Local support for the shared 3.3 V network.
1.8 V / 8	4.7 μ F: C25–C28. 0.47 μ F: C29–C32.	Local support for the FPGA auxiliary supply network.
1.0 V / 10	100 μ F: C34. 4.7 μ F: C35–C39. 0.47 μ F: C40–C43.	Bulk and local bypass support for core/block-RAM power.



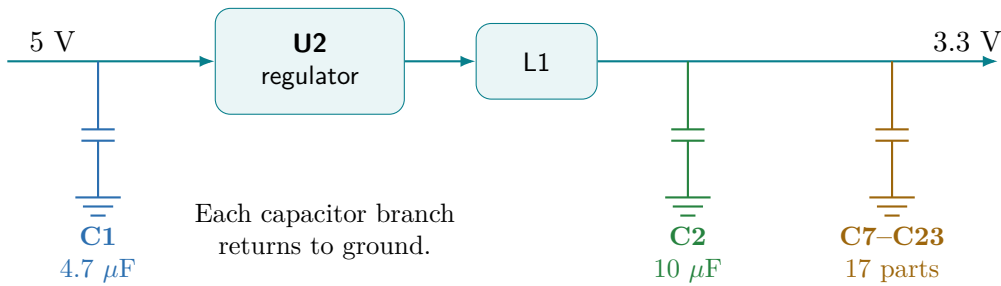
These parts sit underneath and around the FPGA. Putting capacitors near the chip helps shorten the current path, but effective decoupling also depends on vias, traces and planes. Capacitors on the same rail share the supply job; the netlist alone does not identify one specific FPGA pin or logic block served by each part.

Left: original GitHub underside render [E]. Above: direct underside placement drawing, mirrored from source coordinates. Together, the two maps label all **46 capacitors**: 16 top + 30 bottom.

A small charge reserve beside the circuit

A capacitor stores electrical charge. When a circuit suddenly needs current, a nearby capacitor can supply part of it while the power source responds. This helps limit brief voltage dips and ripple. Its **value, package, connections and placement** all matter.

Example: the existing 3.3 V supply



Simplified functional diagram of the audited circuit. C1 = 4.7 μF; C2 = 10 μF. The C7–C23 branch represents 17 distributed capacitors, not one part. Feedback circuitry is omitted here. [E]

Where it is	What to remember
Before the regulator	C1 helps supply switching-current pulses at the input. C3 and C5 do the corresponding job for U3 and U4.
After the inductor	C2 and L1 help smooth the output; C2 also supports changing load current. C4/L2 and C6/L3 serve the other two rails.
Around the FPGA	C7–C43 provide distributed supply support. Larger 47/100 μF parts provide bulk charge; 4.7/0.47 μF parts contribute local bypassing. Their combined electrical behavior depends on the layout.

Same value does not mean same job. C15 and C44 are both 0.47 μF. C15 connects 3.3 V to ground. C44 connects **U2’s feedback pin** to ground, acting on the voltage-sensing/control node. The intended behavior of C44–C46 remains an open design question; do not treat them as spare FPGA bypass capacitors.

When shrinking the board: keep the required power behavior, then evaluate smaller packages or a revised capacitor network. Removing a part because another has the same value or shares the same rail is not enough justification. Pages 15–17 give every exact value; page 23 gives the catalog identities and ratings.